Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Team name: FreeWill

Team leader: Atishay Srivastava

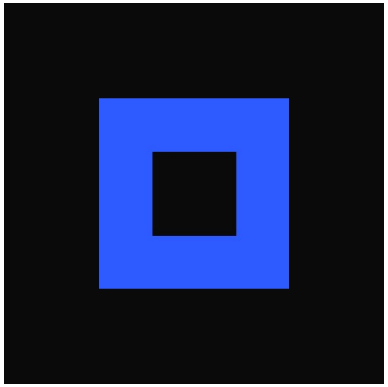
Problem statement

Neighbourhood shopping needs a trustworthy path from a request and private price agreement to payment proof and one physical handover.

Brief about the idea

A shopper scans a shop QR and asks in plain language.

Two personal agents clarify the request, negotiate within each owner’s private limit, and ask both people to confirm the exact deal.



A simulated payment creates a pickup pass. The seller can collect it once, while anyone with the receipt link can verify Collab’s signature.

Opportunity and difference

Retail gap

Discovery and chat do not tell a shop whether a negotiated order was paid or already handed over.

Collab’s approach

Each side runs an A2A agent. Policy limits what can be shared; code rejects offers outside the owner’s limit.

Retail proof

Signed test payment, public receipt verification and seller-only, one-time pickup connect the digital deal to the counter.

Features in the working prototype

Scan and ask

A shop QR opens its agent; Gemini helps draft the shopper’s request.

Discover and negotiate

A2A agents ask questions and make price moves inside private limits.

Check a price

Owner-only Google Search evidence, with a five-check daily cap.

Confirm and pay

Both owners approve; agents sign cart and payment mandates in test mode.

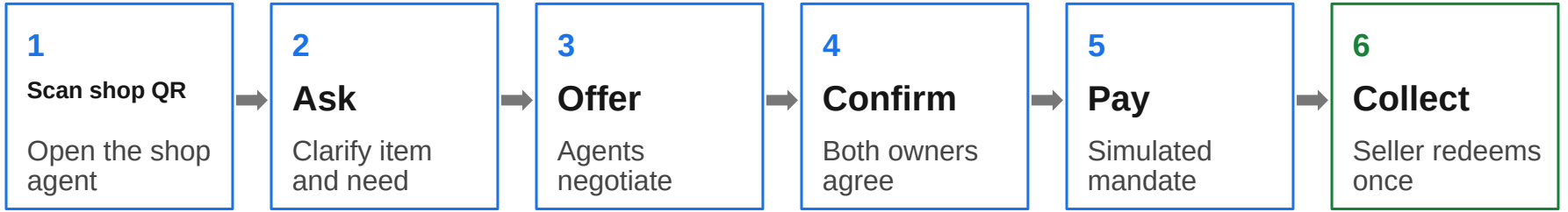
Collect once

Buyer QR shows paid status; only the seller can mark it collected.

Verify publicly

Logged-out receipt page checks Ed25519 signatures against published keys.

Retail purchase process



The receipt binds the item, amount, people and agreement time. A second pickup scan displays the original collection time.

Shopper and counter wireframes

COLLAB REQUESTS

Steel Bottle Store

Water bottles · ₹410 each

✓ **Paid (test mode)**

PICKUP PASS

View signed receipt

PICKUP PASS

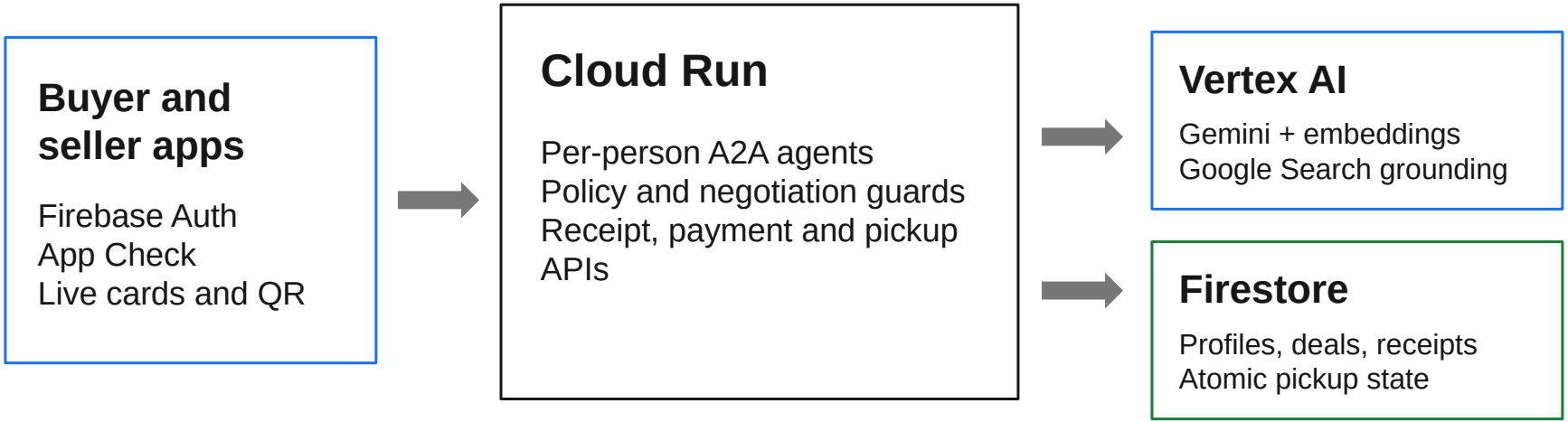
✓ **Paid · Not collected yet**

Water bottles
INR 410
Buyer: Anu

MARK COLLECTED

Editable schematic of the current UI. Payment is simulated.

Solution architecture



Firebase Hosting and Cloud Messaging serve the app; Cloud Scheduler runs due watches; Secret Manager holds signing keys.

Google Cloud and protocol stack

Gemini on Vertex AI	Intent, request drafting, matching, answers and price negotiation
Search grounding	Private price evidence with source links or abstention
Cloud Run + A2A SDK	One agent endpoint per person and the retail API
Firestore	Vector search, deal records, receipt registry and pickup state
Firebase	Hosting, Auth, App Check and Cloud Messaging
Scheduler + Secret Manager	Recurring watches and stable Ed25519 signing key

Trust built in: fraud prevention

Pickup/booking pass	Second scan refused; the seller is matched by account
Signed receipts	Ed25519-signed; any edit fails. Demo: “Try changing the price”
Model Armor	Screens every message from other agents for prompt injection
No overselling	Per-buyer limits and atomic stock holds
Seller record	Public record counts problems from signed receipts
App Check	Checks every write; unverified browsers are rate-limited
Demo payments	Clearly tagged test mode; no real money moves

Prototype cost evidence

$1.10

Estimated Gemini token cost for the full local benchmark

1,402 Gemini calls

This is a token estimate, not a cloud bill. Google Search grounding, Cloud Run, Firestore and networking are excluded. Production cost depends on usage.

Production prototype evidence

Paid listing-origin order on the deployed app, 15 Oct 2026

Buyer’s deal card

Steel Bottle Store
Water bottles · ₹410 each
✓ Paid in test mode
Pickup QR available

Public pickup page

✓ Paid · Not collected yet
Item: Water bottles
Amount: INR 410
Buyer: Anu

Schematic reconstruction of the live screens. The public receipt page verified a Collab-issued Ed25519 attestation; this older order’s payment signatures are legacy HMAC.

Prototype performance and tests

58 / 58
automated tests passed

38 / 39
feasible price pairs agreed

100 / 100
canary probes withheld

Intent routing on real Gemini: p50 1.1 s, p95 1.4 s (120 requests).

Every agreed benchmark price stayed within both parties' private limits. Ten infeasible pairs made no deal.

Benchmark used one product, repeated intent and attack templates, an in-memory store and laptop-to-Vertex latency. All 100 privacy probes escalated to the owner.

Next steps and provenance

Near-term additions

Pickup-window tradeoffs; one basket for several items; photo search for buyers; block and report; export my data.

Later roadmap

Real payment-provider integration and UPI reconciliation. The hackathon prototype moves no money.

Provenance: reading, matching and negotiation logic was ported from the authors' native Collab app. The web app and retail workflow were built for this event.

Product and submission links

Public GitHub repository

https://github.com/kirav5/collab

Demo video (3–4 minutes)

Add link after recording. Required by the submission form.

Deployed prototype

https://collab-55.web.app

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Thank you

